A Multi-Dimensional Knowledge Graph Framework for Relationship Analysis, Contradiction Detection and Research Gap Discovery in Scientific Literature

AUTHOR ONE

DEPARTMENT, INSTITUTION NAME, CITY, POSTAL CODE, COUNTRY.

your.email@institution.edu

Abstract. Reading a new subfield means holding thirty papers in mind at once, and the things a researcher most needs to know are rarely stated in any one of them: which papers are really addressing the same question, where two groups report different numbers for what appears to be the same experiment, and what the collection as a whole has never examined. We present a pipeline that recovers these cross-document relationships automatically and, importantly, makes every answer traceable to the sentence that produced it. Papers are segmented by relative font size rather than by matching heading text, which survives the layout variation of real PDFs; typed entities are extracted by rule; and each pair of documents is scored on five interpretable dimensions whose weights sum to one, so a score can always be decomposed. Disagreements are found by comparing reported measurements and by natural language inference, and are confirmed only when the two papers are methodologically comparable. On a corpus of thirty arXiv papers arranged in four topical clusters, segmentation succeeded on all thirty documents and recovered 90% of expected sections; relationship ranking reached 80% precision@10 against a 23% random baseline. Contradiction detection returned no false positives on this corpus, though it also returned no confirmed contradictions, and we examine why. The system requires no language model API and runs offline.

Keywords: Knowledge graphs, Scientific literature mining, Information extraction, Contradiction detection, Research gap discovery, Document similarity

1 Introduction

A researcher entering an unfamiliar subfield reads somewhere between twenty and forty papers before its structure becomes clear to them. Much of what they learn along the way is not written down in any of those papers. It exists only in the relationships between them.

Four of these relationships matter more than the rest. The first is simple relatedness, which is harder than keyword overlap suggests: two groups can attack an identical problem in vocabulary that shares almost nothing, while two papers stuffed with

the same terminology turn out to be unrelated. The second is disagreement. When one paper reports 76.4% top-1 accuracy for a baseline and another reports 71.1% for what appears to be the same architecture on the same dataset, that discrepancy is a fact about the literature, and it is invisible unless both numbers are in view simultaneously. The third is absence. A research gap is defined by what is missing, and no amount of careful reading of what is present will reveal it; the whole collection has to be surveyed before the hole becomes visible. The fourth is structure, the mental map that an experienced reader of a field carries around, which is a graph whether or not anyone draws it.

Large language models can be asked about all four, and they will answer fluently. The difficulty is that the answer arrives without provenance. A reviewer cannot check it, a student cannot learn from it, and a wrong answer is indistinguishable in tone from a right one. We took the opposite approach. Every number this system produces decomposes into named components, and every contradiction and gap it reports carries the document, section, page and exact sentence that gave rise to it. A language model appears in the pipeline only at the end, and only to phrase findings that have already been established by other means.

Our contributions are as follows.

- A segmentation method for research PDFs built on relative font size, with a guaranteed fallback, which recovered structure from all thirty documents in our evaluation corpus.
- A five-dimensional relationship score with weights constrained to sum to one, which keeps the composite interpretable and lets any ranking be re-derived under different assumptions.
- A two-detector approach to contradiction detection, combining numeric comparison of reported measurements with natural language inference, gated by a methodological comparability test.
- An evaluation corpus of thirty arXiv papers deliberately structured into four clusters so that both true positives and negative controls exist by construction, released with the system.

2 Related Work

2.1 Document representation for scientific text

Sentence-BERT [1] made dense sentence embeddings cheap enough to compute over a whole corpus, and SciBERT [2] showed that pretraining on scientific text rather than general web text improves nearly every downstream task in this domain. SPECTER [3] is closest in spirit to our scoring component: it learns document-level representations using the citation graph as a training signal, so that papers citing one another land near each other in vector space. These methods produce a single vector per document. That is their strength for retrieval and their weakness for our purpose, because a single number expressing how close two papers are cannot say whether

they are close because they share a dataset, a method, or an objective. We keep the dimensions separate for exactly that reason.

2.2 Information extraction from papers

SciERC [4] annotates scientific abstracts with entities and relations and trains a multi-task model over them, and DocRED [5] extends relation extraction to the document level where most relations in practice live. Both require supervision we do not have for the entity types we care about, and both operate on text that someone has already segmented. We found that the segmentation step, which these papers reasonably treat as solved, was the single largest source of failure in our own system.

2.3 Claim verification and contradiction

FEVER [6] established the task of verifying a claim against retrieved evidence, and SciFact [7] adapted it to scientific abstracts, labelling claims as supported, refuted or lacking information. These are supervised tasks with annotated claim-evidence pairs. Our setting is harder in one respect and easier in another: we have no labels, but we also do not need to verify a claim against the world, only to notice when two papers in the same collection cannot both be right. Natural language inference models trained on MultiNLI [8] transfer reasonably to this, which is why DeBERTa [9] appears in our pipeline, but on sentences dense with figures they are unreliable, and a numeric comparison is both cheaper and more trustworthy.

2.4 Citation analysis

Bibliographic coupling [10] and co-citation [11] predate all of the above by several decades and remain useful. Two papers that cite the same prior work are related even when they share no vocabulary at all, which is precisely the signal that embedding-based similarity misses. We include it as one of our five dimensions, though at a lower weight, for reasons discussed in Section 3.

3 Method

The pipeline runs eleven stages in a fixed order. The ordering is forced by data dependency rather than chosen: three of the five scoring dimensions are cosine similarities between entity vectors, so embedding has to complete before scoring begins. We learned this the hard way. In an early version the embedding stage silently failed, and because a missing vector yields a similarity of zero rather than an error, the composite score quietly collapsed to its dataset-overlap term while continuing to produce plausible-looking output.

3.1 Recovering document structure

A PDF stores positioned glyphs and nothing about sections. Recovering the structure turned out to be the hardest part of this work, and it is worth describing the attempts that failed, since the obvious approaches all do.

Matching heading text against a list of section names fails because authors do not agree on the names: Background, Prior Work, Literature Review and Related Literature all denote the same section. Looking for bold text fails because headings in many templates are not bold. Looking for short lines fails badly in two-column layouts, where every line is short. Our first working version combined these heuristics and produced zero sections on one paper in the corpus and one hundred and eighty-six on another, from identical code.

What does work is relative font size. We compute the body text size for each document individually as the character-weighted mode of font sizes on the page, which came to 10.0pt for most papers in our corpus. Any line set noticeably larger, typically 12.0pt, becomes a heading candidate. Because the threshold is relative to the document rather than absolute, it transfers across publishers with different house styles. Candidates are then matched against anchored patterns covering the synonym sets above, after repairing a common extraction artefact in which small-capital headings emerge with a space after the first letter, so that I NTRODUCTION must be read as INTRODUCTION.

No document is discarded. When heading detection finds nothing at all, the paper is divided into three-page chunks labelled as unstructured body text and continues through every subsequent stage. Which route a document took is recorded on the document itself, so a degraded result is visible rather than silent.

3.2 Entity extraction and the missing-section problem

Entities are extracted by rule: gazetteers for datasets, metrics and model names, and cue-phrase patterns for objectives, limitations and explicitly stated gaps. A sentence containing both a metric mention and a number becomes a quantitative result. This needs no training data and no API key, which was a design requirement rather than a convenience, since the system must run on unpublished manuscripts that cannot be sent to a third party.

The section a sentence comes from is evidence about what it contains. An objective is overwhelmingly likely to appear in an abstract, introduction or conclusion, and we use that: each entity type carries an affinity set of sections it may be drawn from. This raises an obvious problem for documents that could not be segmented, where positional evidence does not exist. Our answer is to suspend the restriction entirely for unstructured segments. The trade is deliberate and asymmetric. Such a document loses precision, because its entities are no longer corroborated by position, but it keeps recall and still contributes objectives, methods and results to the analysis.

A third case remains, where a paper genuinely contains no methodology: an editorial, a position piece, an extended abstract. Here there is nothing to recover and we do not pretend otherwise. The methodology dimension evaluates to zero and the weights

are not renormalised over the surviving dimensions. Renormalising is tempting and we rejected it, because it would make a paper matching on a single dimension indistinguishable in score from one matching on all five. The per-dimension scores accompany every composite, so a reader can see which evidence actually existed.

3.3 Five-dimensional relationship scoring

Every pair of documents is scored, giving 435 pairs for a thirty-paper corpus. Each pair is measured on five dimensions that each return a value in [0, 1], combined by a weighted sum as shown in Table 1.

Table 1. The five scoring dimensions, their weights, and the measure used for each.

Dimension	Wt.	Measure	Compares
Objective	0.25	Cosine	Mean embedding of objective statements
Methodology	0.25	Cosine	Mean embedding of method entities
Dataset	0.20	Jaccard	Sets of dataset names
Results and metrics	0.20	Cosine	Reported results and evaluation metrics
Citation	0.10	Jaccard	Reference lists (bibliographic coupling)

The three semantic dimensions compare mean entity embeddings by cosine similarity. The dataset dimension does not: dataset names are exact tokens, so a set-overlap measure is more appropriate than a distributional one, and we use the Jaccard index. Citation overlap is bibliographic coupling over parsed reference lists, with a direct citation between the two documents treated as maximal.

The weights are a stated prior and not a learned parameter, which we think is defensible but want to be explicit about. They encode the judgement that what a paper is trying to do and how it does it are jointly what make it that paper, hence half the mass between them; that datasets and metrics are concrete, checkable common ground, hence a further 0.40; and that shared citations are genuine but lagging evidence, since a paper cannot cite work published after it and reference lists are the noisiest region of a PDF to parse. Two properties keep this from being arbitrary. The configuration is validated at startup to sum to exactly 1.0 and the system refuses to run otherwise, which guarantees the composite stays within [0, 1] and remains comparable across runs. And the weights are adjustable at runtime, so any ranking can be recomputed under a different prior without touching code.

One implementation detail cost us several days and is worth recording. Cosine similarity ranges over [-1, 1], but relatedness does not; two documents pointing in opposite directions in vector space are unrelated rather than negatively related. Before we clamped the value, negative similarities were rejected by the data model's range constraint, and 203 of 435 pairs were being dropped from the results without any error being raised. The fix is a single line. Finding it required counting the output rather than reading the code.

3.4 Contradiction detection

Two detectors run and their verdicts are merged, because they fail in different places. The numeric detector parses the values two papers report, verifies that they measure the same metric on the same dataset under conditions that do not conflict, and flags a material difference. It uses no model at all. The inference detector presents two claim sentences to a cross-encoder as a genuine sentence pair and reads off the stance.

Flagging is cheap and being right is not, so confirmation requires four conditions to hold at once: the stance is contradiction, confidence is at least 0.60, the two papers share a dataset or have comparable metrics, and their methodology similarity is at least 0.30. Everything else is reported as unconfirmed rather than suppressed. The methodology condition is what eliminates the dominant false positive, in which two papers report different numbers for the straightforward reason that they measured different things.

Candidate pairs are restricted to documents that already share a bridge concept or score above a relatedness threshold. Unrelated papers cannot contradict one another, and the restriction keeps this stage from becoming quadratic in corpus size.

3.5 Research gap discovery

Three kinds of absence are detected and they are not treated as equally credible. A stated gap is one an author declares outright, and phrases such as "to the best of our knowledge" or "remains unexplored" score above a vague gesture toward future work. A weak-connection gap arises where several papers share a specific concept yet score poorly against one another otherwise, suggesting a subject approached from disconnected directions. An absent pairing is a combination of two concepts that no paper in the corpus has made.

Absent pairings are the weakest evidence by a wide margin, because most pairs of concepts have never been combined and most of them never should be. We therefore require that both concept communities contain at least three papers, that they do not already overlap, and that the two groups are themselves related; if the papers using X have nothing to do with the papers using Y, the fact that nobody combined them carries no information. At most one such gap can reach the final list.

An early version of this stage returned around sixty gaps. That is not a findings list but a haystack, and a reader skims it and trusts none of it. Ranking candidates by importance and returning at most five above a high floor produced output that a person will actually read, which we regard as the more useful behaviour even though it lowers recall.

4 Evaluation Corpus

A randomly drawn corpus cannot evaluate this system. Contradiction detection only fires when two papers report the same metric on the same dataset, and a relationship score can only be judged if the corpus contains both pairs that ought to rank highly and pairs that ought not. A random sample of thirty papers supplies almost no shared

benchmarks and no negative control, and against such a corpus a scorer that rated every pair as highly related would appear to perform well.

We therefore built the corpus as four deliberate clusters of real arXiv papers, shown in Table 2. Cluster membership is the ground truth: two papers in the same cluster address the same problem on overlapping benchmarks and should score as related, while cluster A against cluster D shares virtually nothing and serves as the negative control. Clusters A and C overlap on dataset but differ on method, which tests whether the score distinguishes "same benchmark" from "same approach". The papers cite one another densely, and several report differing values for identical baselines, which is a well-documented phenomenon in this literature rather than something we introduced.

Table 2. The evaluation corpus. Cluster membership is the ground truth for relationship ranking.

Cluster	Theme	n	Role in the evaluation
A	Convolutional image classification	8	Dense cross-citation; same architectures reported with differing baselines
B	Pretrained language models	8	Shared GLUE and SQuAD benchmarks
C	Vision transformers and self-supervised vision	8	Shares ImageNet with A but not method
D	Retrieval, information extraction, knowledge graphs	6	Negative control against cluster A

5 Results

5.1 Segmentation

All thirty documents were segmented using detected headings, with none falling back to page chunking. Across the eight expected section types, 215 of 240 were recovered, a recall of 90%. Recovery was perfect for introductions and reference lists and weakest for results sections at 21 of 30, which is unsurprising: results are frequently folded into a section headed Experiments, or distributed across several subsections with names like Ablations and Comparison that our patterns map elsewhere.

5.2 Relationship ranking

Of the 435 scored pairs, 99 fall within a cluster and 336 across clusters, so a system ranking pairs at random would be correct 23% of the time. Table 3 reports precision at several cut-offs.

Table 3. Relationship ranking precision at increasing cut-offs. A random ranking scores 23%.

Cut-off	Correct	Precision
precision@5	3 / 5	60%
precision@10	8 / 10	80%
precision@20	17 / 20	85%
precision@30	23 / 30	77%
Random baseline	—	23%

Mean within-cluster score was 0.463 against 0.331 across clusters, a separation of 0.132. The separation matters more than the precision figures, because it indicates that the score is discriminative rather than merely correctly ordered at the top.

The two errors in the top ten are instructive rather than embarrassing. Both pair a self-supervised vision paper from cluster C with a convolutional architecture paper from cluster A, and both are arguably correct: MoCo and ResNet genuinely are closely related, since MoCo evaluates its representations on ImageNet using a ResNet backbone. Our cluster assignment treats them as different because one studies architecture and the other studies pretraining. The score is picking up a real relationship that our ground truth, by design, calls a miss.

5.3 Contradictions and gaps

On the arXiv corpus the system confirmed no contradictions. It also produced no false positives. We report both halves of that result because the second is meaningful only alongside the first, and because an earlier configuration confirmed 1,454 contradictions on the same corpus, essentially all of them spurious. The successive gates described in Section 3.4 brought that figure to 79, then to 10, then to zero as each gate was added.

Whether zero is the right answer we cannot say from this corpus alone. To check that recall had not simply been destroyed, we constructed a six-paper synthetic corpus containing two deliberately planted contradictions, each a pair of papers reporting materially different values for the same metric on the same dataset. Both are detected and confirmed, and this is asserted by the automated test suite. The honest reading is that the confirmation rule is calibrated conservatively, and that real cross-paper numeric disagreement in this particular literature is usually accompanied by differences in experimental condition that the detector correctly declines to call a contradiction.

Gap discovery returned five gaps from 1,037 extracted entities and 92 bridge concepts. The resulting knowledge graph contains 1,159 nodes and 2,047 edges. We do not report precision for gaps, because no ground truth exists and constructing one would require an expert to label thousands of candidate pairings. Reporting an unvalidated number would be worse than reporting none, so the gaps are presented with their evidence trails for human judgement.

5.4 Runtime

A full analysis of thirty papers completes in approximately 33 seconds on a four-core CPU with no GPU and no network access, once the embedding model is resident. Embedding dominates at roughly half the total. Enabling the inference-based contradiction detector increases this substantially, and we found it necessary to batch the cross-encoder calls and cap the batch size, since the unbatched implementation issued one model call per claim pair and exhausted CPU memory on a loaded machine.

6 Limitations

The weights are asserted rather than learned. Learning them would require a labelled relatedness dataset for this corpus, which does not exist, and we regard the interpretability of the current scheme as partial compensation rather than a full answer.

Extraction is rule-based and English-only. Gazetteers transfer poorly to fields whose dataset and metric names we have not enumerated, and a paper in another language would segment correctly while yielding almost no entities. Scanned image-only PDFs produce no extractable text; these are reported as ingestion errors rather than skipped silently, but no OCR is attempted.

Pair scoring is quadratic in corpus size. At thirty papers this is irrelevant, and at a few hundred it remains tractable, but the approach as described would not scale to a corpus of tens of thousands without approximate nearest-neighbour retrieval to prune the candidate set first.

Finally, the contradiction result reported above is a single data point from a single corpus in a single field. The gates are tuned conservatively and we have shown they preserve recall on planted contradictions, but we have not shown what their recall is on real ones, because we have no labelled set of real ones to measure against.

7 Conclusion

We have described a system that reads a folder of research papers and reports how they relate, where they disagree and what they collectively omit, with every claim traceable to a specific sentence on a specific page. Segmentation by relative font size handled all thirty documents in our evaluation corpus and recovered 90% of expected sections. The five-dimensional score reached 80% precision@10 against a 23% baseline while remaining fully decomposable. Contradiction detection produced no false positives, and we have been explicit that it also produced no confirmed findings on real papers.

The design commitment throughout was that an answer a reader cannot verify is not worth giving. That constraint ruled out several easier architectures and cost us accuracy in places. We think it is the right trade for a tool intended to be used on literature a researcher will then go and read for themselves.

The implementation, the thirty-paper corpus and the evaluation scripts are available so that the figures above can be reproduced.

8 References

1. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In: Proceedings of EMNLP-IJCNLP, pp. 3982–3992 (2019).
2. Beltagy, I., Lo, K., Cohan, A.: SciBERT: A Pretrained Language Model for Scientific Text. In: Proceedings of EMNLP-IJCNLP, pp. 3615–3620 (2019).
3. Cohan, A., Feldman, S., Beltagy, I., Downey, D., Weld, D.S.: SPECTER: Document-level Representation Learning using Citation-informed Transformers. In: Proceedings of ACL, pp. 2270–2282 (2020).
4. Luan, Y., He, L., Ostendorf, M., Hajishirzi, H.: Multi-Task Identification of Entities, Relations, and Coreference for Scientific Knowledge Graph Construction. In: Proceedings of EMNLP, pp. 3219–3232 (2018).
5. Yao, Y., Ye, D., Li, P., Han, X., Lin, Y., Liu, Z., Sun, M.: DocRED: A Large-Scale Document-Level Relation Extraction Dataset. In: Proceedings of ACL, pp. 764–777 (2019).
6. Thorne, J., Vlachos, A., Christodoulopoulos, C., Mittal, A.: FEVER: a Large-scale Dataset for Fact Extraction and VERification. In: Proceedings of NAACL-HLT, pp. 809–819 (2018).
7. Wadden, D., Lin, S., Lo, K., Wang, L.L., van Zuylen, M., Cohan, A., Hajishirzi, H.: Fact or Fiction: Verifying Scientific Claims. In: Proceedings of EMNLP, pp. 7534–7550 (2020).
8. Williams, A., Nangia, N., Bowman, S.: A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference. In: Proceedings of NAACL-HLT, pp. 1112–1122 (2018).
9. He, P., Liu, X., Gao, J., Chen, W.: DeBERTa: Decoding-enhanced BERT with Disentangled Attention. In: Proceedings of ICLR (2021).
10. Kessler, M.M.: Bibliographic coupling between scientific papers. American Documentation 14(1), 10–25 (1963).
11. Small, H.: Co-citation in the scientific literature: A new measure of the relationship between two documents. Journal of the American Society for Information Science 24(4), 265–269 (1973).
12. Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In: Proceedings of NAACL-HLT, pp. 4171–4186 (2019).
13. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention Is All You Need. In: Advances in Neural Information Processing Systems 30, pp. 5998–6008 (2017).
14. He, K., Zhang, X., Ren, S., Sun, J.: Deep Residual Learning for Image Recognition. In: Proceedings of CVPR, pp. 770–778 (2016).
15. He, K., Fan, H., Wu, Y., Xie, S., Girshick, R.: Momentum Contrast for Unsupervised Visual Representation Learning. In: Proceedings of CVPR, pp. 9729–9738 (2020).
16. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In: Advances in Neural Information Processing Systems 33, pp. 9459–9474 (2020).
17. Ji, S., Pan, S., Cambria, E., Marttinen, P., Yu, P.S.: A Survey on Knowledge Graphs: Representation, Acquisition, and Applications. IEEE Transactions on Neural Networks and Learning Systems 33(2), 494–514 (2022).

18. Hagberg, A.A., Schult, D.A., Swart, P.J.: Exploring Network Structure, Dynamics, and Function using NetworkX. In: Proceedings of the 7th Python in Science Conference, pp. 11–15 (2008).